Chapter 1

$\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$

The set $\mathbb{R}^n$

- $\mathbb{R}$ is the set of all real numbers.
- Let n be a positive integer.
- We can write $(x_1, x_2, \cdots, x_n)$ of real numbers, which is an ordered n-tuple of real numbers.
- $\mathbb{R}^n$ is the set of all ordered n-tuples of real numbers.
- An element $x \in \mathbb{R}^n$ can be written in

$$\text{row} \quad x = (x_1, x_2, x_3, \cdots, x_n) \quad \text{or in column} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix}, \quad \text{or any other form.}$$

  As long as the order of listed n real numbers are seen, there would not be a problem.

Addition and Scalar multiplication

Addition

1. If $x = (x_1, x_2, \cdots, x_n) \in \mathbb{R}^n$ and $y = (y_1, y_2, \cdots, y_n) \in \mathbb{R}^n$, we can add the two to obtain an element in $\mathbb{R}^n$:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \\ \vdots \\ x_n + y_n \end{pmatrix},$$

   which is denoted by $x + y \in \mathbb{R}^n$.

Scalar multiplication

1. In this course, the scalar is a synonym of real number.
2. If λ is a real number and $x = (x_1, x_2, \cdots, x_n) \in \mathbb{R}^n$, then we can scale x to obtain λx:

$$\lambda \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \\ \lambda x_3 \\ \vdots \\ \lambda x_n \end{pmatrix} \in \mathbb{R}^n.$$

The set $\mathbb{R}^{m\times n}$

- Let $\mathbb{R}$ be again the set of all real numbers.
- Let m and n be two positive integers.
- The set of all $(A_{ij})_{i=1,j=1}^{i=m,j=n}$ of real numbers indexed by $(i,j) \in \{1,2,\cdots,m\} \times \{1,2,\cdots,n\}$ is denoted by $\mathbb{R}^{m\times n}$.
- The notation for an element $A \in \mathbb{R}^{m\times n}$ in this time is more specific. We distinguish two indices $i = 1,2,\cdots,m$ and $j = 1,2,\cdots,n$. We list mn real numbers in a box so that i is a row index, and j is a column index:

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \cdots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \cdots & A_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1} & A_{m2} & A_{m3} & \cdots & A_{mn} \end{pmatrix}.$$

- We call an element of $\mathbb{R}^{m\times n}$ an $(m \times n)$ matrix, which reads as "m by n matrix".
- Why ..?

Addition and Scalar multiplication

Addition

1. If $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times n}$, i.e., if the matrix A and B are in same shape, we can add the two to obtain an element in $\mathbb{R}^{m\times n}$:

$$\begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \cdots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \cdots & A_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1} & A_{m2} & A_{m3} & \cdots & A_{mn} \end{pmatrix} + \begin{pmatrix} B_{11} & B_{12} & B_{13} & \cdots & B_{1n} \\ B_{21} & B_{22} & B_{23} & \cdots & B_{2n} \\ B_{31} & B_{32} & B_{33} & \cdots & B_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ B_{m1} & B_{m2} & B_{m3} & \cdots & B_{mn} \end{pmatrix}$$
$$= \begin{pmatrix} A_{11}+B_{11} & A_{12}+B_{12} & A_{13}+B_{13} & \cdots & A_{1n}+B_{1n} \\ A_{21}+B_{21} & A_{22}+B_{22} & A_{23}+B_{23} & \cdots & A_{2n}+B_{2n} \\ A_{31}+B_{31} & A_{32}+B_{32} & A_{33}+B_{33} & \cdots & A_{3n}+B_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1}+B_{m1} & A_{m2}+B_{m2} & A_{m3}+B_{m3} & \cdots & A_{mn}+B_{mn} \end{pmatrix},$$

   which is denoted by $A + B \in \mathbb{R}^{m\times n}$.

Scalar multiplication

1. If λ is a real number and $A \in \mathbb{R}^{m\times n}$, then we can scale A to obtain λA:

$$\lambda \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \cdots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \cdots & A_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1} & A_{m2} & A_{m3} & \cdots & A_{mn} \end{pmatrix} = \begin{pmatrix} \lambda A_{11} & \lambda A_{12} & \lambda A_{13} & \cdots & \lambda A_{1n} \\ \lambda A_{21} & \lambda A_{22} & \lambda A_{23} & \cdots & \lambda A_{2n} \\ \lambda A_{31} & \lambda A_{32} & \lambda A_{33} & \cdots & \lambda A_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \lambda A_{m1} & \lambda A_{m2} & \lambda A_{m3} & \cdots & \lambda A_{mn} \end{pmatrix}.$$

In summary,

1. We have the set $\mathbb{R}^n$, equipped with the addition and the scalar multiplication, that we denote by $\big(\mathbb{R}^n, +, s\big)$.

$$+ : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n,$$
$$s : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n.$$

2. We havethe set $\mathbb{R}^{m\times n}$, equipped with the addition and the scalar multiplication, that we denote by $\big(\mathbb{R}^{m\times n}, +, s\big)$.

$$+ : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n,$$
$$s : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n.$$

1. That we work with $\big(\mathbb{R}^n, +, s\big)$, i.e., that $\mathbb{R}^n$ are equipped with the addition and the scalar multiplication may, more importantly, mean that <u>we do not perform other operations</u>.

   These are illegal expressions:

   (a) $\mathbb{R}^n + \mathbb{R}$, $n \geq 2$:

   (b) $\mathbb{R}^n + \mathbb{R}^m$, $n \neq m$.

   (c) product in general. $\times : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$, though in some dimensions we can define meaningful product.

   (d) comparison in general.

2. Likewise, for the case of $\mathbb{R}^{m\times n}$, <u>we do not perform other operations</u>, unless otherwise defined later of this course.

- We have so far
$$(\mathbb{R}^n, +, s) \quad \text{and} \quad (\mathbb{R}^{m\times n}, +, s),$$
respectively a set equipped with the addition and scalar multiplication.

- Expression that makes use of the addition and scalar multiplication, and use only of them such as
$$\lambda_1 x + \lambda_2 y + \lambda_3 z + \lambda_4 w, \quad x, y, z, w \in \mathbb{R}^n$$
comes as an important expression to us.

- The expression is called a "Linear Combination" of x, y, z, w; Linear, meaning that the scalar multiplication $x \mapsto \lambda x$ is to make something proportional, and Combination, meaning here we are adding things.

The final comment on $(\mathbb{R}^n, +, s)$ and $(\mathbb{R}^{m\times n}, +, x)$:

As stated earlier, in some dimensions we may additionally define product, for example in $\mathbb{R}^2$ we can define

$$(x_1, x_2) \times (y_1, y_2) = (v_1, v_2), \quad \text{where} \quad v_1 + iv_2 = (x_1 + ix_2)(y_1 + iy_2).$$

However, all of such operations are treated in this course to be exceptional and come as addendum, emphasizing that there are only <u>(i) the addition</u> and <u>(ii) the scalar multiplication</u> we are allowed to operate so far.

Chapter 2

Matrix Matrix multiplication

- Emphasizing that there are no other general product operations, now we define the unique product operation between matrices.
- For given matrix $A \in \mathbb{R}^{\ell \times m}$ and matrix $B \in \mathbb{R}^{m \times n}$, we define the matrix $C \in \mathbb{R}^{\ell \times n}$ to be the product denoted by AB so that

$$C_{ij} = \sum_{\alpha=1}^{m} A_{i\alpha} B_{\alpha j} \quad \text{for } i = 1, 2, \cdots, \ell \text{ and } j = 1, 2, \cdots, n.$$

- Importantly, the product is defined only for the case where the second component of shape of A and the first component of shape of B are the same, and the product is not defined for all remaining cases.
- Why .. ?

The defined product can be regarded in a few different ways.

1. The product can be regarded as to build new n linear combinations of columns of A, as designated by numbers in B:

2. The product can be regarded as to build new ℓ linear combinations of rows of B, as designated by numbers in A:

3. The product can be regarded as to build new ℓn real numbers, out of rows of A and columns of B.

The product is useful in many places.

Perfect for the chain rule in multivariable calculus

Suppose that $X = \mathbb{R}^n$, $Y = \mathbb{R}^m$, and $Z = \mathbb{R}^\ell$, and

$$f : X \to Y, \quad g : Y \to Z.$$

Multivariable Calculus:

1. Also assume f and g are many times diffentiable functions.
2. For a given point $\bar{x}$, we collect numbers
$$B_{\alpha j} = \frac{\partial f^\alpha}{\partial x^j}(\bar{x}) = \lim_{h \to 0} \frac{f^\alpha(\bar{x} + he_j) - f^\alpha(\bar{x})}{h}.$$
3. Similarly, for a given point $\bar{y}$, we collect numbers
$$A_{i\alpha} = \frac{\partial g^i}{\partial y^\alpha}(\bar{y}) = \lim_{h \to 0} \frac{g^i(\bar{y} + he_\alpha) - g^i(\bar{y})}{h}.$$
4. Now we consider the composition
$$g \circ f : X \to Z, \quad g \circ f(x) = g(f(x))$$
and collect numbers
$$C_{ij} = \frac{\partial (g \circ f)^i}{\partial x^j}(\bar{x}) = \lim_{h \to 0} \frac{(g \circ f)^i(\bar{x} + he_j) - (g \circ f)^i(\bar{x})}{h}.$$
5. Then C turns out to equal to AB, for $\bar{y} = f(\bar{x})$.

Writing a system of linear equations

If we are given

$$\begin{cases} 3x - 7y + 4z = -2, \\ 9x - 2y - 6z = 0, \\ -5x + 3y - 11z = -8 \end{cases}$$

It is nice that we can write also

$$\begin{pmatrix} 3 & -7 & 4 \\ 9 & -2 & -6 \\ -5 & 3 & -11 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ -8 \end{pmatrix}$$

Using $\ell = 3$, $m = 3$, and $n = 1$ in the left-hand-side.

The question of why..?

1. Why does it have to be in that way?
2. Why not we define for $A, B \in \mathbb{R}^{m\times n}$ a product with the same shape in $\mathbb{R}^{m\times n}$?

$$\begin{pmatrix} A_{11}B_{11} & A_{12}B_{12} & A_{13}B_{13} & \cdots & A_{1n}B_{1n} \\ A_{21}B_{21} & A_{22}B_{22} & A_{23}B_{23} & \cdots & A_{2n}B_{2n} \\ A_{31}B_{31} & A_{32}B_{32} & A_{33}B_{33} & \cdots & A_{3n}B_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1}B_{m1} & A_{m2}B_{m2} & A_{m3}B_{m3} & \cdots & A_{mn}B_{mn} \end{pmatrix}$$

   (a) Actually you can. In most of computer language, this entry-wise product in the same shape is provided by the broadcasting.

3. All why-questions come to the following one,

   Would other people be interested in the new definition ?

A few important observations

1. Important: You don't mess up with the order of multiplication here.
2. Unlike multiplication we know for two real numbers, for the Matrix-Matrix multiplication for $A \in \mathbb{R}^{\ell \times m}$ and $B \in \mathbb{R}^{m \times n}$, the product BA is not even defined in general. This is simply because in general $n \neq \ell$.
3. In the special case of that $\ell = m = n$, BA is defined. However even in such a case

$$AB \neq BA \quad \text{in general.}$$

Summary

1. We have $\left(\mathbb{R}^{m \times n}, +, s\right)$ for m and n positive integers.
2. Linear combinations of members of $\mathbb{R}^{m \times n}$ are expressions using $+$ and s such as

$$\lambda_1 A_1 + \lambda_2 A_2 + \cdots \lambda_j A_j \in \mathbb{R}^{m \times n}.$$

3. We have a definition of multiplication for $A \in \mathbb{R}^{\ell \times m}$ and matrix $B \in \mathbb{R}^{m \times n}$ resulting in $AB \in \mathbb{R}^{\ell \times n}$.

Knowledge so far will be applied to mathematics, science, and engineering, and will be extremely important and powerful.

Chapter 3

Linear Independence and Basis

Let us consider elements in $\mathbb{R}^2$,

$$x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \quad z = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

It is easy to notice that if we are with x and y in our hands, z can be reproduced by

$$z = 2\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \end{pmatrix},$$

a linear combination of x and y.

Linear Combinations and Span

For a given finite number of elements $a_1, a_2, \cdots, a_k \in \mathbb{R}^n$, the set of all linear combinations of them is denoted by

$$\mathsf{span}\left\langle a_1, a_2, \cdots, a_k \right\rangle = \left\{ x \in \mathbb{R}^n \mid x \text{ is a linear combination of } a_1, a_2, \cdots, a_k \right\}.$$

Example

Let

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix} \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix} \quad \cdots \quad e_n = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

Then, $e_1, e_2, \cdots, e_n$ span all elements in $\mathbb{R}^n$, i.e., $\mathsf{span}\left\langle e_1, e_2, \cdots, e_n \right\rangle = \mathbb{R}^n$.

Linear Independence

As far as reproducibility is concerned, we notice that we keep only two elements

$$x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ 2 \end{pmatrix},$$

and every element in $\mathbb{R}^2$ can be reproduced by linear combination of x and y.

This necessiates the notion of linear independece of a few elements in $\mathbb{R}^n$.

Example 1 z above is linearly dependent with respect to x and y.

Example 2 On the other hand, in $\mathbb{R}^3$,

$$x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}, \quad z = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

Then, x, y, z are linearly independent. By no chance, z is a linear combination of x and y.

Definition 1. *We say $a_1, a_2, \cdots, a_k \in \mathbb{R}^n$ are linearly independent if*

$$\lambda_1 a_1 + \lambda_2 a_2 + \cdots + \lambda_k a_k = 0 \quad \Longrightarrow \quad \lambda_1 = \lambda_2 = \cdots = \lambda_k = 0.$$

$a_1, a_2, \cdots, a_k \in \mathbb{R}^n$ are said to be linearly dependent if they are not linearly independent.

Basis

Definition 2. *We say $a_1, a_2 \cdots, a_k$ of $\mathbb{R}^n$ form a basis of $\mathbb{R}^n$ if*

1. *$a_1, a_2 \cdots, a_k$ span $\mathbb{R}^n$,*
2. *$a_1, a_2 \cdots, a_k$ are linearly independent.*

Examples

$e_1, e_2, \cdots, e_n$ form a basis.

We will prove in later section the following seemingly natural facts:

1. Elements in $\mathbb{R}^n$ fewer than n cannot span <u>all</u> of $\mathbb{R}^n$.
2. Elements in $\mathbb{R}^n$ exceeding n must linearly <u>dependent</u>.
3. n elements in $\mathbb{R}^n$ are linearly independent if and only if they <u>do span all</u> of $\mathbb{R}^n$.

Chapter 4

Power and Importance of $\mathbb{R}^{m\times n}$

We saw that $\left(\mathbb{R}^{m\times n}, +, s\right)$ is so simply defined. The power of $\mathbb{R}^{m\times n}$ comes from that, (i) there are so many instances in mathematics, science, engineering, and etc., where entities of interests are such simple that we can add and scale them and that (ii) we will learn nice tools for $\mathbb{R}^{m\times n}$.

Example 1 In fact, this is an example of $\mathbb{C}^{m\times n}$, with $\mathbb{C}$ of complex numbers.
Wolfgang Pauli, a physicist, introduced *Pauli matrices* in developing theory of quantum mechanics in early 20c.

We consider a multiplication table for a 4 elements set $\{1, \sigma_x, \sigma_y, \sigma_z\}$:

Table 4.1: A multiplicative table

	1	σ_x	σ_y	σ_z
1	1	σ_x	σ_y	σ_z
σ_x	σ_x	-1	σ_z	$-\sigma_y$
σ_y	σ_y	$-\sigma_z$	-1	σ_x
σ_z	σ_z	σ_y	$-\sigma_x$	-1

Note that $\sigma_x\sigma_y \neq \sigma_y\sigma_x$, and this set cannot be represented by elements in $\mathbb{R}$.

One of the way we achieve the table:

We consider
$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \right\}$$
for $1, \sigma_x, \sigma_y, \sigma_z$ from the left.

The set $\mathsf{span}\left\langle \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \right\rangle$ contains all the elements that can be generated.

note: The role played by $\mathbb{R}$, the scalar factor, can be taken by the mathematical object of *field*. But we will stick to $\mathbb{R}$ in this course.

Example 2 Colors (Light)

Although there are so many colors such as yellow, green, blue, white, black, $\cdots$, it turns out that the only three of them, red, green, and blue are independent. All others are linear combination of the three.

We choose $[0, 255]^3 \subset \mathbb{R}^3$ represents the colors, with

1. $(255, 0, 0) \simeq$ red,
2. $(0, 255, 0) \simeq$ green,
3. $(0, 0, 255) \simeq$ blue,
4. $(0, 0, 0) \simeq$ black,
5. $(255, 255, 255) \simeq$ white.

The value ranged in $[0, 255]$ represents the intensity.

Example 3 Bacteria with manipulatable genes.

In Bio-Engineering, ceartain bacateria is called a *vector*. This reflects that a certain kind of bacteria may or may not contain a few genes of our interests, and can be manipulated in the laboratory in the way of addition.

By a reproduction, a bacteria with gene A and a bacteria with gene B can result in a bacteria with both genes A and B. The number of such genes that can be maniplulated in lab. and bacteria can be represented by $\mathbb{R}^n$.

Example 4 Sauce

Sauce can be perfectly represented by $\mathbb{R}^n$. They can be added and scaled:

We linear combinate the sugar, salt, vinegar, soy sauce, water, what you want, in grams.

So a recipe you have to remember for a certain sauce is an element in $\mathbb{R}^n$, and if you want to make sauce a lot, you scale the element.

1. We saw many examples of *entities* that can be added and scaled, and sometimes we employ multiplication.
2. They are well-represented by $\mathbb{R}^{m\times n}$ or $\mathbb{R}^n$.
3. In all of examples, we see that there are two different roles:

$$\text{Respresentation} \quad \text{vs.} \quad \text{Essential Entities}$$

4. The theory has been developed in Linear Algebra so that the two roles are explicitly detached from each other. The latter, of essential entities, is defined in mathematics as *vector space* and *algebra*.
5. In our course, we use the word *vector* to denote the essential entity, while we simply use the word element of $\mathbb{R}^n$, or the matrix in $\mathbb{R}^{m\times n}$ for the representational use.
6. The connection from the set of essential entities to for instance $\mathbb{R}^n$, is done by fixing the basis. Let us be clear on this below.

- The definition of vector space X over the scalar $\mathbb{R}$ can be found in math textbooks and webpages. For completeness, we also contained the mathematical definition in the last page of this chapter.
- If you are not familiar with mathematical definitions, consider a vector space X as a set of symbols that can be added and scaled.
- k-dimensional vector space for k a positive integer.

  Definition 3. *A vector space X that admits a basis consists of k elements is called a k-dimensional vector space.*

  Example The space of colors admits a basis of three elements. Hence the space is 3-dimensional vector space.

- Connection from the *essential entities* to *representing elements* is done once we fix what are the basis elements in X.

  Example The space of sauces consists of $a = sugar$, $b = salt$, $c = vinegar$, and $d = water$.

  Once we fix a, b, c, and d to be the basis, we will understand an element

$$\begin{pmatrix} 5 \\ 5 \\ 3 \\ 100 \end{pmatrix} \in \mathbb{R}^4$$

  to represent a sauce consisting of 5-sugar, 5-salt, 3-vinegar, and 100 water, in grams.

  We may want to take the diluted ingredients $a + 100d$, $b + 100d$, $c + 100d$, and d to be the basis. Then

$$\begin{pmatrix} 5 \\ 5 \\ 3 \\ 100 \end{pmatrix} \in \mathbb{R}^4$$

  to represent a sauce consisting of 5-sugar, 5-salt, 3-vinegar, and 1400 water, in grams.

- You can see the detached roles of *representation* and *essential entities*.
- And how the basis choice connects the two.

A vector space X over the scalar $\mathbb{R}$ is mathematically defined as follows.

1. X is a nonempty set.
2. $+ : X \times X \to X$ is well-defined with following properties.
   (a) For any $x, y, z \in X$, $x + (y + z) = (x + y) + z$.
   (b) For any $x, y \in X$, $x + y = y + x$.
   (c) $\exists 0 \in X$ such that for any $x \in X$ $x + 0 = x$.
   (d) For any $x \in X$, $\exists(-x) \in X$ such that $x + (-x) = 0$.
3. $s : \mathbb{R} \times X \to X$ is well-defined with follwoing properties.
   (a) For any $\lambda_1, \lambda_2 \in \mathbb{R}$ and $x \in X$, $\lambda_1(\lambda_2 x) = (\lambda_1\lambda_2)x$
   (b) For any $x \in X$, $1x = x$.
4. $+$ and s work in compatible ways satisfying the following.
   (a) For any $\lambda \in \mathbb{R}$ and $x, y \in X$, $\lambda(x + y) = \lambda x + \lambda y$.
   (b) For any $\lambda_1, \lambda_2 \in \mathbb{R}$ and $x \in X$, $(\lambda_1 + \lambda_2)x = \lambda_1 x + \lambda_2 x$.

Summary

1. For an m-dimensional vector space $(X, +, s, \{b_1, b_2, \cdots, b_m\})$ with m basis vectors $b_1, b_2, \cdots, b_m$, we consider the representation of its element for example

$$\begin{pmatrix} 3 \\ -5 \\ 0 \\ \vdots \\ 0.8 \end{pmatrix} \in \mathbb{R}^{m\times 1}$$

   to represent $3b_1 - 5b_2 + 0b_3 + \cdots + 0.8b_m$.

2. Be careful when X itself is $\mathbb{R}^m$, which is the very possible example. One way we will follow in this course is that, for the representational use, we make use of

$$\mathbb{R}^{m\times 1} \quad \text{of one column matrices.}$$

   and try best to distinguish the essential entity X and its representation.

Chapter 5

Transformation of Solid shapes

We consider the three dimensional space we live.

1. We consider the vector space X and algebra L of essential entities below.
2. Their repective representation will be $\mathbb{R}^{3\times 1}$ and $\mathbb{R}^{3\times 3}$.

We do the folloing for solid shapes:

The vector space X

1. We let the set

$$X = \{\text{Locations}\} \quad \text{with respect to a fixed origin.}$$

2. Scalar multiplication: Location with respect to an origin can be scaled by a real number.
3. Addition: take the parallologram law.

Basis selection

1. We take the length 1 orthogonal locations e_1, e_2, and e_3 be the basis of X.
2. Then every location is a linear combination of those three locations.

Representation in $\mathbb{R}^{3\times 1}$ columns

1. The coordinates of a location is then represented in $\mathbb{R}^{3\times 1}$.

A solid shape is then a subset occupying a region of locations.

$$G \subset X$$

Linear Transformations

- A transformation is then a mapping

$$T : X \to X.$$

- We say $T : X \to X$ is linear if

$$\begin{aligned} (i) \quad & T(\lambda x) = \lambda T(x) \\ (ii) \quad & T(x+y) = T(x) + T(y). \end{aligned}$$

  i.e., $T(\lambda_1 x_1 + \lambda_2 x_2 + \cdots + \lambda_k x_k) = \lambda_1 T(x_1) + \lambda_2 T(x_2) + \cdots \lambda_k T(x_k)$.

- We then collect

$$L = \big\{ T : X \to X \;\big|\; T \text{ is linear} \big\}.$$

Basis selection in L

1. There exists a linear mapping $\hat{T}^{1,1}$ such that

$$\hat{T}^{1,1}(e_1) = e_1, \quad \hat{T}^{1,1}(e_\beta) = 0 \quad \beta = 2, 3.$$

2. Similarly there exists a linear mapping $\hat{T}^{i,\alpha}$ such that

$$\hat{T}^{i,\alpha}(e_\alpha) = e_i, \quad \hat{T}^{i,\alpha}(e_\beta) = 0 \quad \beta \neq \alpha.$$

3. The 9 elements for $i = 1, 2, 3$ and $\alpha = 1, 2, 3$ are the basis of L

Representation in $\mathbb{R}^{3\times 3}$

1. With such 9 basis, a linear transformation is represented in $\mathbb{R}^{3\times 3}$.
2. For example

$$\hat{T}^{1,1} \quad \text{is represented by} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Since we have fixed basis and representations, from now on, we work with numbers and matrices.

i.e., we have $x \in \mathbb{R}^{3\times 1}$ and $A \in \mathbb{R}^{3\times 3}$, by which, the new location is represented by

$$y = Ax \in \mathbb{R}^{3\times 1}.$$

Here, we list a few elementary matrices, that does:

Dilation, Rotation, Projection, Reflection, and Shearing.

Identity Matrix

1. We consider the matrix
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \in \quad \mathbb{R}^{3\times 3}$$
2. It is a special element in $\mathbb{R}^{3\times 3}$, and denoted by I.
3. Note that for any $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$,
$$y = Ax = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$
   - Diagonal entries are numbers in diagonal of the square.
   - Off diagonal entries are entries not in diagonal.
   - All of diagonal entries of I are 1
   - All of off diagonal entries of I are 0.
4. Be careful that I is not the one below !
$$I \neq \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Diagonal Matrices

1. Let $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}^3$. We consider a matrix

$$\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \quad \in \quad \mathbb{R}^{3\times 3}.$$

2. First, let us consider the special cases $\lambda_1 > 0, \lambda_2 > 0, \lambda_3 > 0$

   (a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad Ax =$

   (b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad Ax =$

   (c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad Ax =$

   (d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad Ax =$

3. $\lambda_i = 0$: One component is shrinken to 0.
4. $\lambda_i < 0$: Dilate with sign flip.
5. More matrices for the dilations below will be introduced later.

   (a) Principal axes are not coordinate axes.

   (b) Dilation with directions not orthogonal to each other.

Elementary Rotation Matrices

1. Let $\theta \in \mathbb{R}$. We consider a matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \in \quad \mathbb{R}^{3\times 3}.$$

(a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad Ax =$

(b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad Ax =$

(c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \qquad Ax =$

(d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \qquad Ax =$

2. We consider a matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} \quad \in \quad \mathbb{R}^{3\times 3}.$$

(a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad Ax =$

(b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad Ax =$

(c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \qquad Ax =$

(d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \qquad Ax =$

3. We consider a matrix

$$\begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \quad \in \quad \mathbb{R}^{3\times 3}.$$

(a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad Ax =$

(b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad Ax =$

(c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \qquad Ax =$

(d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \qquad Ax =$

Elementary Orthogonal Projection Matrices

1. We consider a matrix
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \in \quad \mathbb{R}^{3\times 3}.$$

   (a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad Ax =$

   (b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad Ax =$

   (c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \qquad Ax =$

   (d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \qquad Ax =$

2. We will introduce more orthogonal projection matrices onto the projecting plane that is not aligned with coordinate.

3. There are non-orthogonal projections, but we will not use them much. We will introduce them later.

Elementary Reflection Matrices

1. We consider a matrix
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad \in \quad \mathbb{R}^{3\times 3}.$$

   (a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad Ax =$

   (b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad Ax =$

   (c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \qquad Ax =$

   (d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \qquad Ax =$

2. We will introduce more reflection matrices whose symmetry hyperplane is not aligned with coordinate.

Upper Triangular matrices: Elementary Shearing

1. We consider a matrix
$$\begin{pmatrix} \lambda_1 & \alpha & \beta \\ 0 & \lambda_2 & \gamma \\ 0 & 0 & \lambda_3 \end{pmatrix} \quad \in \quad \mathbb{R}^{3\times 3}.$$

2. This kind of matrices are called upper triangular matrices.

   (a) Example 1 : $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad Ax =$

   (b) Example 2 : $x = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad Ax =$

   (c) Example 3 : $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \qquad Ax =$

   (d) Example 4 : $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \qquad Ax =$

We can consider a few consecutive applications of linear transformation,

$$T = A_k A_{k-1} A_{k-2} \cdots A_2 A_1$$

of 3×3 matrices.

1. We examined finitely many 3×3 elementary matrices that do:

   dilation, rotation, projection, reflection, shearing

   but only with respect to three axes or to xy, yz, and zx planes.

2. Now, we learn the technique to incorporate those 5 deformations with axes not along the coordinate axes.

3. To be concrete, let us consider a reflection with respect to a plane passing the origin that is not xy, yz, or zx plane.

1. Having seen the plane, we became to think: if I knew the reflection was what I was going to to, I should have selected three locations of the basis adapted to that plane.

2. Now, let us pose another observer who happened to choose the basis as I wished, say the location f_1, f_2, and f_3.

3. He of course use $\mathbb{R}^{n+1}$ to represent the location, but in his language, this is a linear combination of his basis.

$$\begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} \quad \text{refers to the location} \quad 1f_1 - 3f_2 + 4f_3.$$

The only knowledge we need to translate his language

1. What if I want to know the location for

$$\begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}$$

   in my language ?

2. Let us make analogy, saying his language is english, and our language is korean. We would need

   English-to-Korean Dictionary or Korean-to-English Dictionary

3. If the location f_1, f_2, and f_3 are known in my language, for example

$$f_1 = \begin{pmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix}, \quad f_2 = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{pmatrix}, \quad f_3 = \begin{pmatrix} -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ \frac{2\sqrt{6}}{6} \end{pmatrix}$$

   This is going to be the English-to-Korean Dictionary. Let us list

$$F = \begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix}$$

   that is my English-to-Korean Dictionary.

4. If so, another making use of Matrix-Matrix multiplication:

   the location in my language $= \begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix} \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} = 1 \begin{pmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix} - 3 \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{pmatrix} + 4 \begin{pmatrix} -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ \frac{2\sqrt{6}}{6} \end{pmatrix}.$

5. In summary, if I know the English-to-Korean Dictionary $\begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix}$ and I know the location in English $\begin{bmatrix} a' \\ b' \\ c' \end{bmatrix}$ then the location in Korean $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is in relation

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix} \begin{bmatrix} a' \\ b' \\ c' \end{bmatrix}.$$

Now, it is time for Korean-to-English Dictionary

1. Now, I want to write things in English, for given a location in my language, I want to know how it is represented in his language.

2. That is of course,

$$\begin{bmatrix} a' \\ b' \\ c' \end{bmatrix} = \begin{bmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{bmatrix}^{-1} \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

We will re-visit the inverse matrix later to study it properly, but here we just admit that B^{-1} of our matrix B exists with the property we want.

Definition 4. *For a given $A \in \mathbb{R}^{3\times 3}$, if there exists a matrix $L \in \mathbb{R}^{3\times 3}$ such that*

$$AL = LA = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which must be then unique, L is said to be the inverse matrix of A, and denoted by A^{-1}.

---

1. Suppose that I have the Dictionary and suppose I also know its inverse. Then I can refer to any location in both language.
2. Then, if we get back to the reflection problem, we can do:
   (i) Translate the location to his language.
   (ii) Do the elementary reflection we just learned, that is along his axes.
   (iii) Since the reflected result will be in his language, I translate it into my language.

That is to perform, for given $x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$,

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \mapsto \begin{bmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{bmatrix}^{-1} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{bmatrix} a' \\ b' \\ c' \end{bmatrix} \qquad \text{Korean to English}$$

$$\begin{bmatrix} a' \\ b' \\ c' \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} a' \\ b' \\ c' \end{bmatrix} = \begin{bmatrix} d' \\ e' \\ f' \end{bmatrix} \qquad \text{Elementary reflection in his language}$$

$$\begin{bmatrix} d' \\ e' \\ f' \end{bmatrix} \mapsto \begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix} \begin{bmatrix} d' \\ e' \\ f' \end{bmatrix} = \begin{pmatrix} d \\ e \\ f \end{pmatrix} \qquad \text{English to Korean}$$

This is written all at once,

$$\begin{pmatrix} d \\ e \\ f \end{pmatrix} = \begin{pmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{pmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} | & | & | \\ f_1 & f_2 & f_3 \\ | & | & | \end{bmatrix}^{-1} \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

the reflected location in my language.

Chapter 6

Linear mappings from X to Y

- We will re-visit $\mathbb{R}^{n\times n}$ of square matrices, to discuss the determinat of a square matrix, the inverse if exists, how we can calculate them, and so on.
- Continuing discussions on the question: What does the matrix A do ?
- On the matrix $A \in \mathbb{R}^{m\times n}$ that is not square.
- A matrix in $\mathbb{R}^{n\times n}$ is called a square matrix. A matrix in $\mathbb{R}^{m\times n}$ is called a rectangular matrix. Of course, a square matrix is a rectangular matrix.

- Because we have developed the same on : "what does a square matrix A do?",
- We can also easily understand "what does a rectangular matrix do?" too.
- This is done through the notion of the rank of matrix.

First of all, in this chapter,

- $\mathbb{R}^{n\times 1}$ represent a vector space X of dimension n,
- $\mathbb{R}^{m\times 1}$ represent another vector space Y of dimension m,
- $\mathbb{R}^{m\times n}$ represent the space of linear mappings from X to Y.

Rank of a matrix A

Let $A \in \mathbb{R}^{m \times n}$, i.e., it has m rows and n columns, in the way

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \cdots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \cdots & A_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{m1} & A_{m2} & A_{m3} & \cdots & A_{mn} \end{pmatrix}.$$

Motivation and observation: when the number "one" is relevant

Consider a special matrix $A \in \mathbb{R}^{3 \times 5}$,

$$A = \begin{pmatrix} 1 & 2 & 10 & 200 & 3 \\ 2 & 4 & 20 & 400 & 6 \\ 3 & 6 & 30 & 600 & 9 \end{pmatrix}$$

so that every columns are constant multiple of the first column.

But, it is also such that every rows are constant multiple of the first row.

Try differently. Is this always so ?

Motivation and observation: when the number "two" is relevant

Consider a special matrix $A \in \mathbb{R}^{2\times 3}$,

$$A = \begin{pmatrix} 3 & 0 & -9 \\ 1 & 2 & 1 \end{pmatrix}$$

Certainly, $(3, 0, -9)$ and $(1, 2, 1)$ are the two independent rows.

Are the three columns independent ?

No. Certainly, $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ are linearly independent.
However, $\begin{pmatrix} -9 \\ 1 \end{pmatrix} = -3 \begin{pmatrix} 3 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 2 \end{pmatrix}$, i.e., $\begin{pmatrix} -9 \\ 1 \end{pmatrix} \in \mathsf{span} \left\langle \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\rangle$.

Thus, the number of linearly independent columns coincide the number "two".

It is not obvious to see why, but the number of linearly independent rows and the number of linearly independent columns seem to be the same.

To be able to check the statement in mathematical way, we do the followings.

1. We collect
$$\mathsf{span}\left\langle n \text{ columns of } A\right\rangle \subset \mathbb{R}^{m\times 1}.$$

2. And we also collect
$$\mathsf{span}\left\langle m \text{ rows of } A\right\rangle \subset \mathbb{R}^{1\times n}.$$

The question is how many number of linearly independent elements in them ?

Before we proceed, we introduce two notations.

- For $A \in \mathbb{R}^{m\times n}$, A^T, the transpose of A, is the matrix in $\mathbb{R}^{n\times m}$ where the rows and columns are interchanged, i.e.,
$$(A^T)_{ij} = A_{ji}, \quad j = 1, 2, \cdots, m, \quad i = 1, 2, \cdots, n.$$

- For two columns $x, y \in \mathbb{R}^{\ell\times 1}$, we say x and y are orthogonal to each other if
$$x^T y = 0.$$

We now give a name for the above two we collected.

1. We call the first one, as
$$\mathrm{Ran}\,(A), \quad \text{the range of } A$$

2. We consider the *transpose* the second one,
$$\mathrm{Trow}(A) = \mathsf{span}\left\langle m \text{ transpose of rows of } A\right\rangle \subset \mathbb{R}^{n\times 1}.$$

The range Ran (A) is literally the range of A.

- Check: if $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, then according to the Matrix Multiplication rule,

  $Ax = x_1c_1 + x_2c_2 + \cdots x_nc_n$, a linear combination of columns of A.

- Here, $c_1, c_2, \cdots, c_n$ are columns of A.

A subset $E \subset X$ is called a vector subspace of X if E is also a vector space.

Ran $(A) \subset \mathbb{R}^{m\times 1}$ is a *vector subspace* of $\mathbb{R}^{m\times 1}$, and Trow$(A) \in \mathbb{R}^{n\times 1}$ is a *vector subspace* of $\mathbb{R}^{n\times 1}$.

The objective of the next lecture is to reveal the following:

1. Trow(A) and Ran (A) are both r-dimensional, and
2. "The question we posed" can be understood by looking at the square matrix of $\mathbb{R}^{r\times r}$

   $$\text{Trow}(A) \to \text{Ran}\,(A).$$

   between the two.
3. What does the square matrix do? We completely know.

This is known as the Fundamental Theorem of Linear Algebra. We will go through this in the next lecture in detail.

Chapter 7

Fundamental Theorem of Linear Algebra

Here, we verify the picture:

through the fundamental theorem of linear algebra. We divide it into two parts.

We recall the orthogonality.

Theorem 1. *Let* $x_0 \in \mathbb{R}^{n\times 1}$.

$$x_0 \perp \mathit{Trow}(A) \quad \Longleftrightarrow \quad Ax_0 = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Proof.

$$\begin{aligned} x_0 \perp \mathrm{Trow}\,(A) \quad &\Longleftrightarrow \quad x_0^T\big(\lambda_1 b_1 + \lambda_2 b_2 + \cdots + \lambda_m b_m\big) = 0, \quad b_1,\, b_2,\, \cdots,\, b_m \text{ are transpose of rows of } A \\ &\Longleftrightarrow \quad x_0^T b_i = 0 \quad \text{for every } i = 1, 2, \cdots, m \\ &\Longleftrightarrow \quad i\text{-th component of } Ax_0 = 0 \text{ for every } i = 1, 2, \cdots, m, \end{aligned}$$

according to the Matrix Multiplication Rule. □

Theorem 2. *For any matrix $A \in \mathbb{R}^{m\times n}$,*

the number of lin. indep. elements in Trow(A) = the number of lin. indep. elements in Ran(A).

- In fact we only need to check that for every matrix $A \in \mathbb{R}^{m\times n}$,

  the number of lin. indep. elements in $\mathrm{Trow}\,(A)$ $\leq$ the number of lin. indep. elements in $\mathrm{Ran}\,(A)$.

- This is because, if that is true, then do the same on another matrix $B = A^T$,

  the number of lin. indep. elements in $\mathrm{Trow}\,(A^T)$ $\leq$ the number of lin. indep. elements in $\mathrm{Ran}\,(A^T)$.

Proof. We only check the inequality:

1. To do that: Suppose $b_1, b_2, \cdots, b_r \in \mathrm{Trow}\,(A)$ are linearly independent. We show that $a_1 = Ab_1$, $a_2 = Ab_2$, $\cdots$, $a_r = Ab_r$ are linearly independent.
2. Method by proving contrapositive statement.
3. Suppose that $a_1, a_2, \cdots a_r$ are linearly dependent. i.e., we can find coefficients λ_1, λ_2, $\cdots$, λ_r, not all of them are zero, such that
$$\lambda_1 a_1 + \lambda_2 a_2 + \cdots + \lambda_r a_r = 0.$$
4. In other words, by linearity,
$$A\big(\lambda_1 b_1 + \lambda_2 b_2 + \cdots \lambda_r b_r\big) = 0.$$
5. But Theorem 1 says that then
$$x_0 = \lambda_1 b_1 + \lambda_2 b_2 + \cdots \lambda_r b_r \in \mathrm{Trow}\,(A) \quad \text{is orthogonal to all of } \mathrm{Trow}\,(A).$$
6. The only element in $\mathrm{Trow}\,(A)$ that is orthogonal to all of $\mathrm{Trow}\,(A)$ is only
$$x_0 = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0. \end{pmatrix}.$$
7. This concludes that $b_1, b_2, \cdots, b_r \in \mathrm{Trow}\,(A)$ are linearly dependent.

$\square$